

Juan Antonio Rodríguez de la Cruz

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Data Scientist Jr. / Master in Artificial Intelligence

Professional Profile

Junior Data Scientist and Biomedical Researcher with Master's in AI (Cum Laude) and PhD candidate in Biomedicine & Molecular Biotechnology. Expertise in curating, annotating, and validating multi-omics datasets for AI/ML training in molecular biology and bioinformatics. Proven in developing DL models (PyTorch) for biomedical image synthesis, reducing processing times 90% via cutting-edge tools. Strong communicator of scientific findings to technical/non-technical stakeholders.

Technical Skills

- **Scientific Data Curation/Annotation:** Multi-Omics, Biomedical Images, Genomic Metadata.
- **AI/ML Frameworks:** PyTorch, Scikit-learn, LSTM Networks, Evolutionary Deep Learning.
- **Data Processing:** Python (Pandas, NumPy), R/Bioconductor, SQL, Data Wrangling/Mining.
- **Tools/Systems:** Linux, Git, Matplotlib, Bioinformatics Pipelines.
- **Languages:** Python, R, SQL | English (Intermediate), Spanish (Native)

Professional Experience

Instituto Politécnico Nacional | Research Student, CDMX, Mexico. 2023/02 – Present

- **Curated multi-omics datasets** for ancestral cancer risk analysis, **annotating genomic regions** to validate AI/ML hypotheses using R/Bioconductor & Python Data Mining.
- **Developed/evaluated evolutionary Deep Learning models** for biological hypotheses; provided domain-specific feedback yielding robust bioinformatics conclusions (PyTorch, ML/DL).
- Implemented state-of-the-art computational pipelines from peer-reviewed papers, slashing workflow times **90%** without quality loss (Git, Linux, PyTorch).

TripleTen DS Certification | Student, CDMX, Mexico. 2025/08 – Present

- **Data Wrangling:** Extracted/validated telecom & transportation datasets, annotating patterns for revenue optimization (SQL, Python).
- **Statistical Analysis:** Cleaned/analyzed user data for insights; documented methodologies (Pandas, NumPy, Matplotlib).

Universidad Veracruzana | Research Student, Veracruz, Mexico. 2019/08 – 2021/09

- **Curated & annotated biomedical image databases** (thousands of images/metadata), ensuring AI training readiness using Python, Pandas, Linux.
- **Developed advanced DL model** for biomedical image synthesis to augment AI training data; **published in 2 indexed journals** with model performance validation (PyTorch, NumPy).

Education

PhD in Biomedicine and Molecular Biotechnology , Instituto Politécnico Nacional	2023 – Present.
Data Scientist, Certification , TripleTen	2025 – 2026.
Master's in Artificial Intelligence , Universidad Veracruzana (Cum Laude)	2019 – 2021.
Bachelor's in Biotechnology , Universidad Veracruzana (Graduated with Honors by GPA)	2015 – 2019.